1. What is a PFT?
2. How is it performed?
3. What are the different patterns of pulmonary function?
4. How does one interpret a PFT?
5. What is bronchodilator response?
6. What is a methacholine challenge?
7. What is obstructive lung disease, and which diseases are associated?
8. What is restrictive lung disease, and which diseases are associated?
9. Which diseases affect the DLCO?
10. What are the different lung volumes and lung capacities?
11. What are the different types of obstruction, and how to dx?

Source: Course Syllabus

Pulmonary Function Tests

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- Categorize the nature and severity of disease
- Follow the progression of disease
- Measure response to therapy
- Define disability
- Pre-op evaluation
- Screening at risk populations
- Monitoring drug toxicity

Pulmonary Function Tests

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- **Tests of Flow**
- **Tests of Volume**
- **Tests of Diffusion Capacity**
- **Tests of Oxygenation**
- **Tests of Ventilation**

Lung Volumes

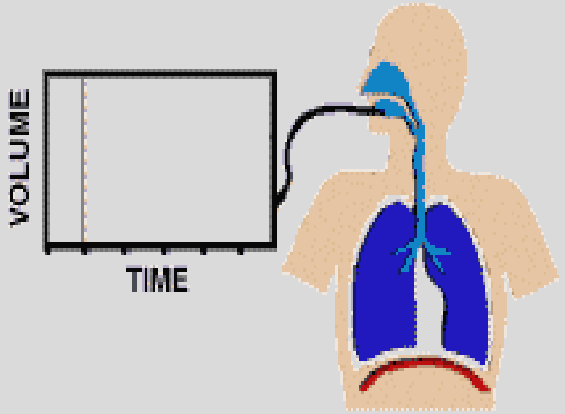
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- IRV= Inspiratory Reserve volume
- V_T= Tidal Volume
- TLC= Total Lung Capacity
- RV= Residual Volume
- ERV= Expiratory Reserve Volume
- IC = Inspiratory Capacity

Spirometry

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- Simple, office-based
- Measures flow, volumes
- Volume vs. Time
- Can determine:
  - Forced expiratory volume in one second (FEV_1)
  - Forced vital capacity (FVC)
  - FEV_1/FVC



Spirometry

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- FVC = Forced Vital Capacity
- FEV_1 = Forced Expiratory Volume in one second
- FEF_{25-75} =
  Forced Expiratory Flow between 75% and 25% of Vital Capacity

Terminology

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- Forced vital capacity (FVC):
  - Total volume of air that can be exhaled forcefully from TLC
  - The majority of FVC can be exhaled in <3 seconds in normal people, but often is much more prolonged in obstructive diseases
  - Measured in liters (L)

Forced Vital Capacity

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- Interpretation of % predicted:
  - 80-120% Normal
  - 50-79% Mild reduction
  - 30%-49% Moderate reduction
  - <30% Severe reduction

Terminology

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- Forced expiratory volume in 1 second: (FEV_1)
  - Volume of air forcefully expired from full inflation (TLC) in the first second
  - Measured in liters (L)
  - Normal people can exhale more than 75-80% of their FVC in the first second; thus the FEV1/FVC can be utilized to characterize lung disease

FEV_1 ---(GOLD Criteria)

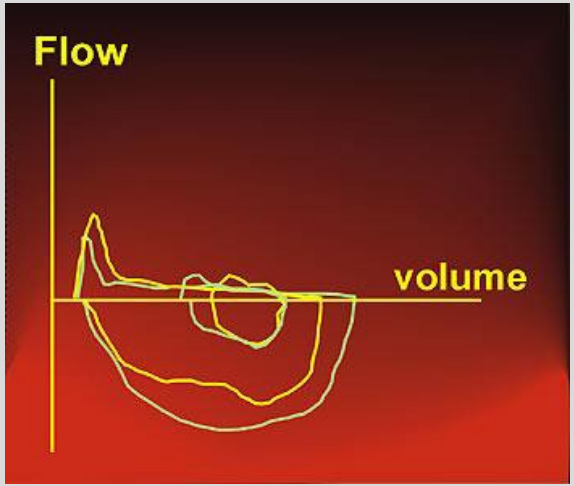
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- In presence of a low FEV_1/FVC ratio:
- Interpretation of % predicted:
  - >=80% Normal
  - >=50 - <80% Moderate obstruction
  - >=30 - <50% Severe obstruction
  - <30% Very Severe obstruction

Obstructive Pattern

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- Decreased FEV_1
- Decreased FVC
- Decreased FEV_1/FVC
  - <70% predicted
- FEV_1 used to follow severity in COPD



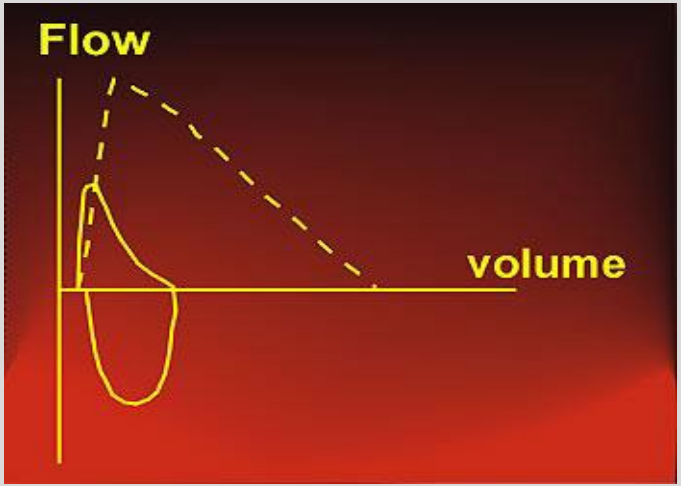
Obstructive Lung Disease — *Differential Diagnosis*

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- Asthma
- COPD
  - chronic bronchitis
  - emphysema
- Bronchiectasis
- Bronchiolitis
- Upper airway obstruction

Restrictive Pattern

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- Decreased FEV_1
- Decreased FVC
- FEV_1/FVC *normal* or *increased*

Restrictive Lung Disease

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Interstitial Lung Disease

- Interstitial pneumonitis
- Pneumoconiosis
- Pulmonary Edema

Space Occupying Lesions

- Tumors
- Cysts

Pleural Diseases

- Pneumothorax
- Hemothorax
- Pleural Effusion
- Fibrothorax

Chest Wall Diseases

- Injury/Flail Chest
- Kyphoscoliosis
- Spondylitis
- Neuromuscular Disease

Restrictive Lung Disease

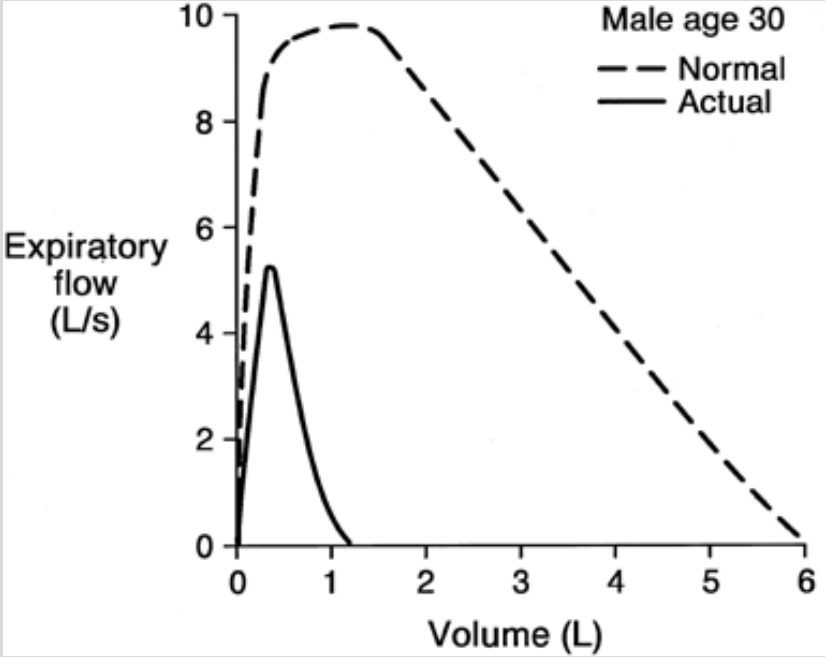
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Extra Thoracic Conditions

- Obesity
- Pregnancy
- Ascites
- Peritonitis

Severe Restrictive Lung Disease

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TLC	2.1 L	(28%)
FVC	1.12 L	(19%)
FEV_1	1.04 L	(21%)
FEV_1/FVC	93%	(100%)
MVV	81 L/min	(43%)
$D_{L_{CO}}$	15 mL/min/mm Hg	(43%)
SaO_2	Rest Exercise	95% 88%
Slope FV curve	7.5	(2-3.0)

Bronchodilator Response

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- Degree to which FEV_1 improves with inhaled bronchodilator
- Documents *reversible* airflow obstruction
- Significant response if:
  - FEV_1 increases by 12% and >200ml
- Request if obstructive pattern on spirometry

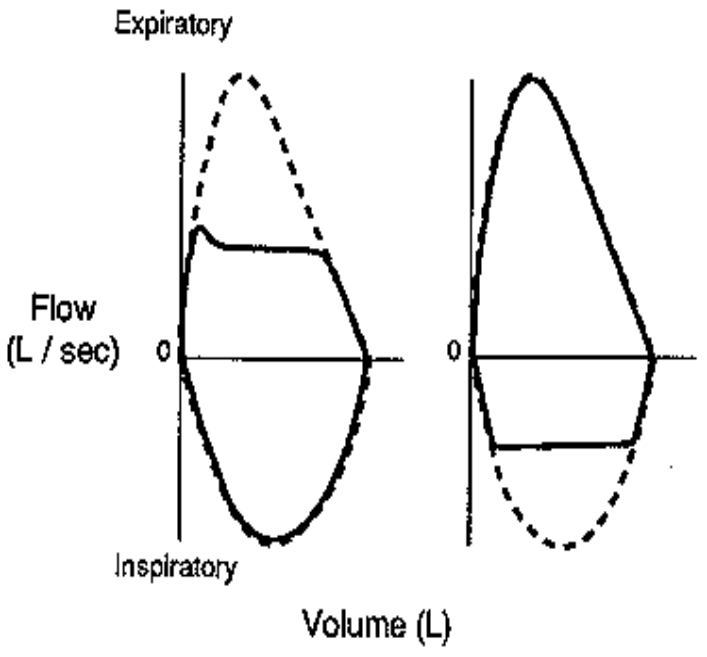
Upper Airway Obstruction

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- Variable intrathoracic obstruction
- Variable extrathoracic obstruction
- Fixed obstruction

Large Airway Obstruction

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- Characterized by a truncated inspiratory or expiratory loop

Flow Volume loops in upper airway obstruction: **Variable *Extra*- thoracic**

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- Edema of epiglottis
- Vocal cord paralysis
- Vocal cord adhesions
- Foreign body

Flow Volume loops in upper airway obstruction: **Variable *Intra*-Thoracic**

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- Polypoid tumor
- Tracheomalacia
- Polychondritis

Flow Volume loops in upper airway obstruction: ***Fixed***

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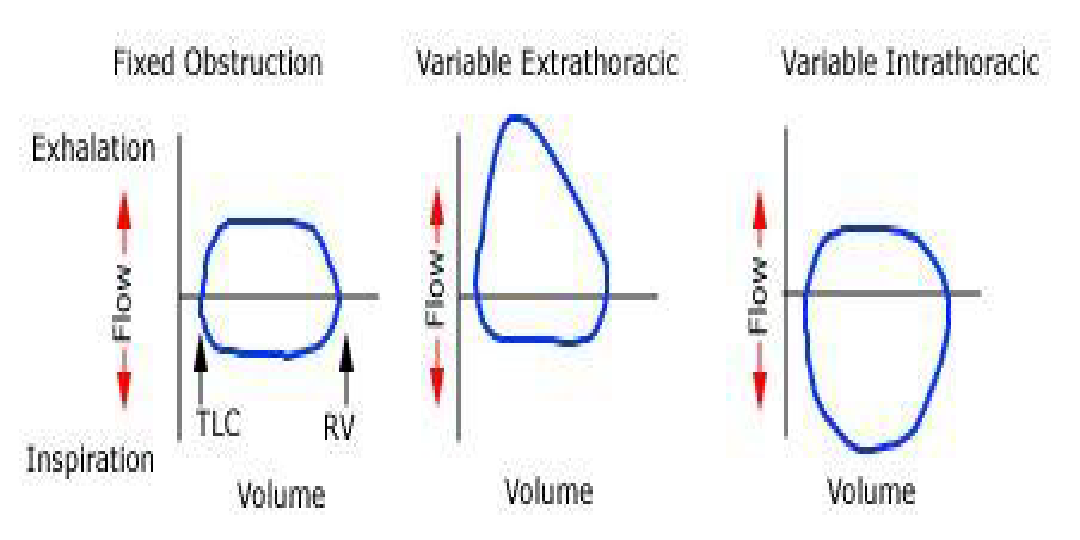
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- Tracheal stenosis
- Tracheal goiter
- Hypertrophied tonsils
- Large foreign body

Upper Airway Obstruction

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Lung Volumes

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- Measurement:
  - helium dilution
  - nitrogen washout
  - body plethysmography
- Indications:
  - Diagnose restrictive component
  - Differentiate types of obstructive lung dz

Lung Volumes

- IRV= Inspiratory Reserve volume
- V_T= Tidal Volume
- TLC= Total Lung Capacity
- RV= Residual Volume
- ERV= Expiratory Reserve Volume
- IC = Inspiratory Capacity

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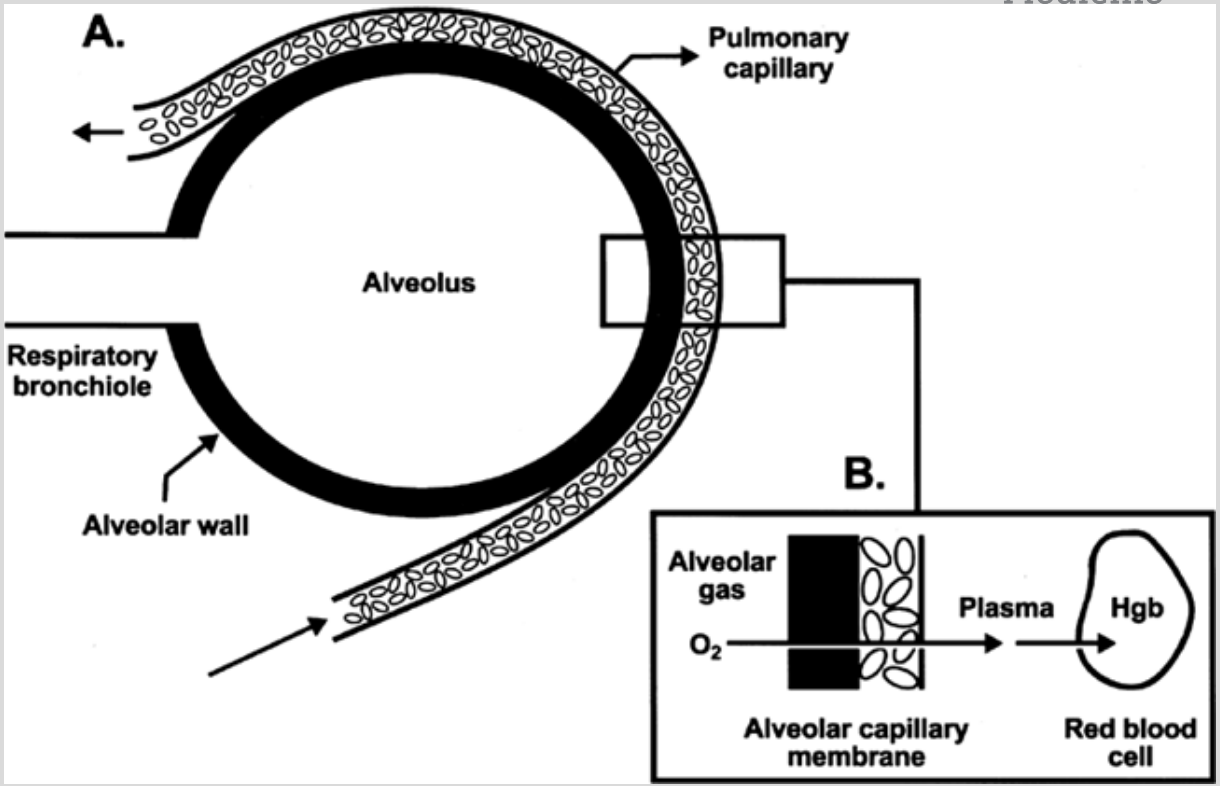
Lung Volumes – Patterns

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- Obstructive
  - **TLC > 120% predicted**
  - RV > 120% predicted
- Restrictive
  - **TLC < 80% predicted**
  - RV < 80% predicted

Diffusing Capacity

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Diffusing Capacity Determined by:

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- Surface area of alveoli
- Thickness of alveolar membrane
- Volume of blood in capillary bed
- Reaction rate of the test gas with Hgb
  - We use **DLCO** – carbon monoxide

Diffusing Capacity

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- **Decreased DLCO**
  (<80% predicted)
  - Obstructive lung disease
    - Emphysema
  - Parenchymal disease
    - Interstitial (fibrosis)
  - Pulmonary vascular disease (pulm HTN)
  - Anemia

- **Increased DLCO**
  (>120-140% predicted)
  - Asthma (or normal)
  - Pulmonary hemorrhage
  - Polycythemia
  - Left to right shunt

DLCO — Indications

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- Differentiate asthma from COPD
- Evaluation and severity of restrictive lung disease
- Early stages of pulmonary hypertension

PFT Patterns

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- **Emphysema**
  - FEV_1/FVC <70%
  - "Scooped" FV curve
  - TLC *increased*
  - Increased compliance
  - DLCO *decreased*
- **Chronic Bronchitis**
  - FEV_1/FVC <70%
  - "Scooped" FV curve
  - TLC *normal*
  - Normal compliance
  - DLCO usually *normal*

Bronchoprovocation

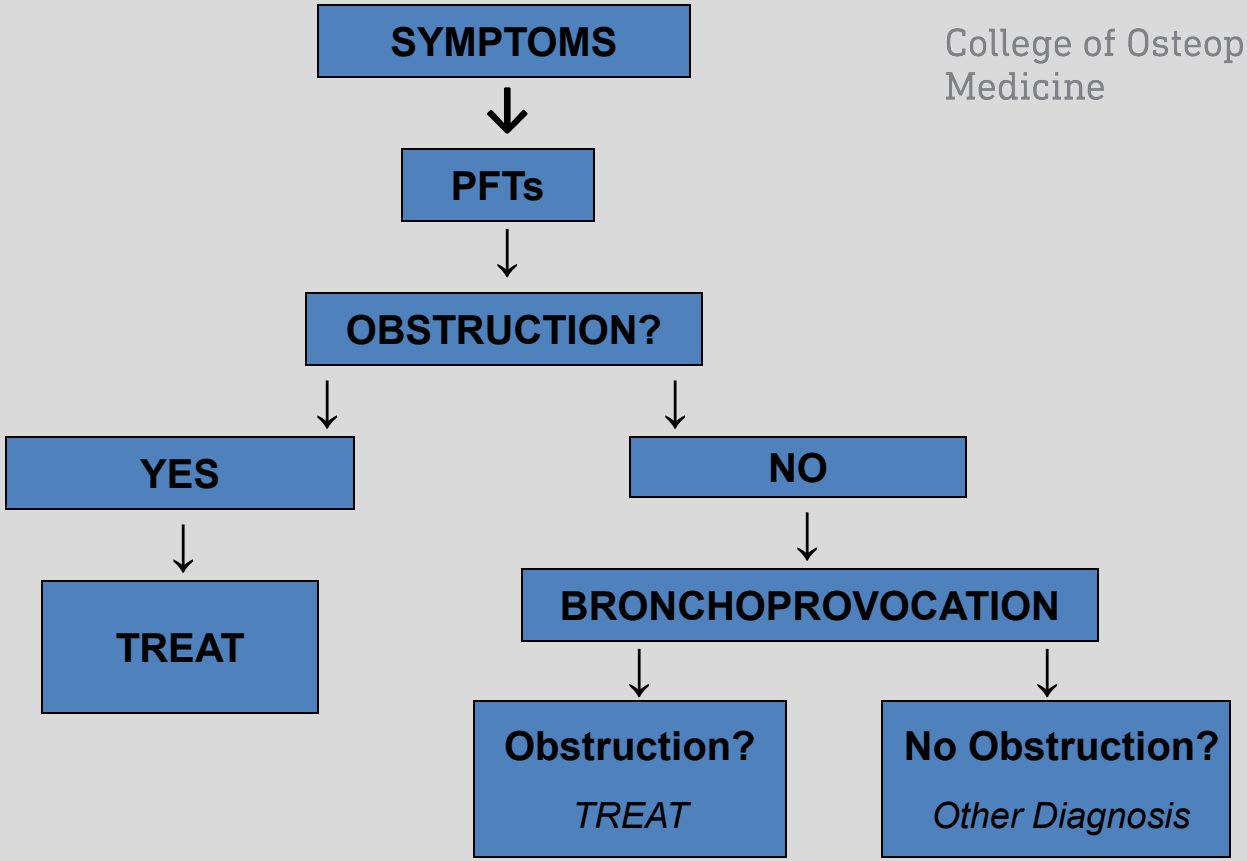
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- Useful for diagnosis of asthma in the setting of *normal* pulmonary function tests
- Common agents:
  - Methacholine, Histamine, Mannitol
- Diagnostic if: ≥20% decrease in FEV_1

Methacholine Challenge

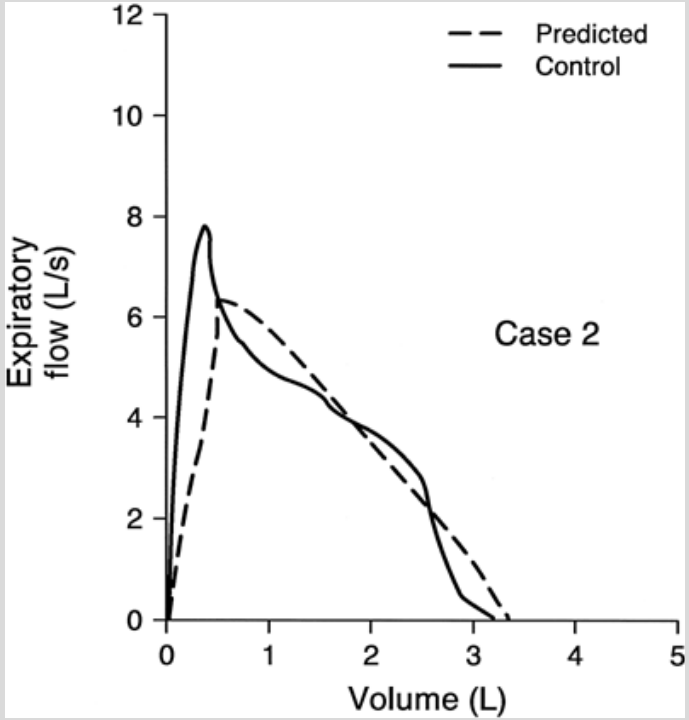
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- Inhaled methacholine
  - Cholinergic drug that stimulates muscarinic receptors
  - Causes contraction of smooth muscle in airway
  - Degree to which muscle reacts depends on its reactivity
- Predictive of asthma attacks by cold air, exercise, pollens, infection, etc.
- Given up to 5 doses


SYMPTOMS
PFTs
OBSTRUCTION?
YES
NO
TREAT
BRONCHOPROVOCATION
Obstruction?
TREAT
No Obstruction?
Other Diagnosis

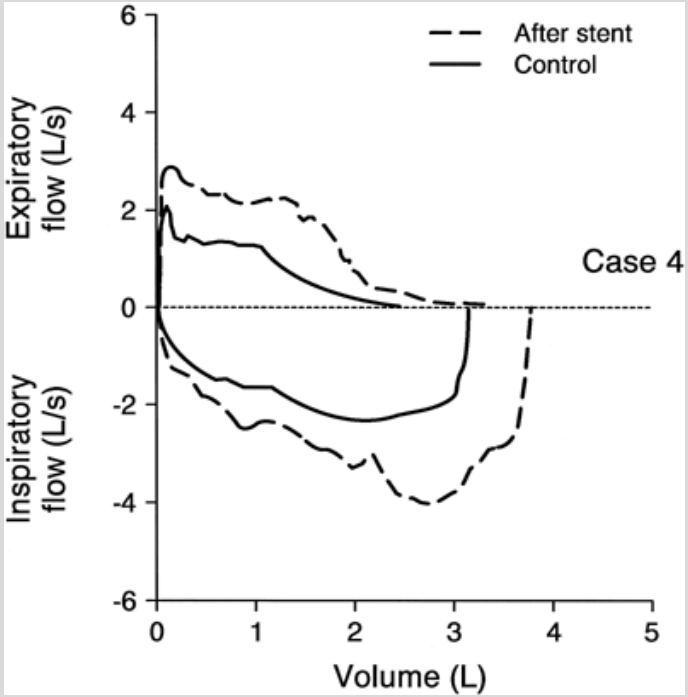
36 y o female

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- Nrml FEV1 ratio
- Nrml FEV1
- Nrml TLC

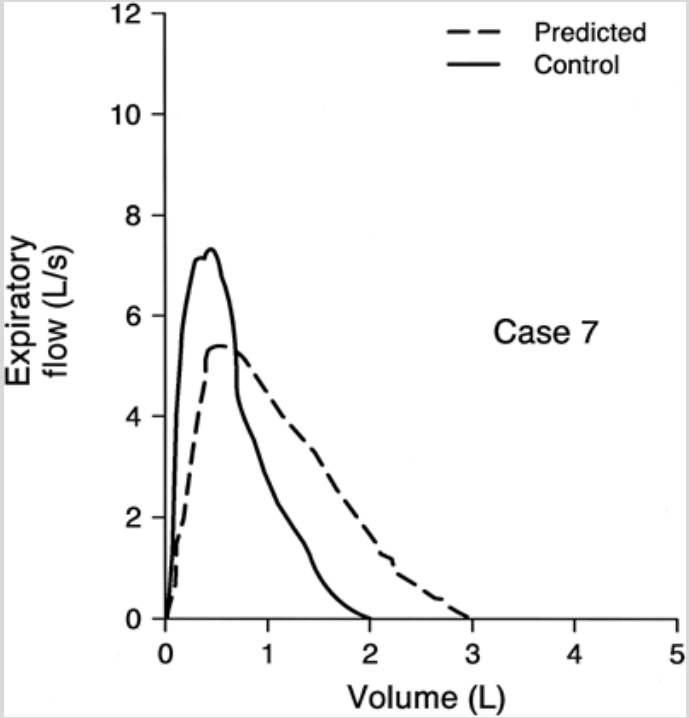
76 y o male - sob



- Decreased FEV1 ratio
- FEV1 47%
- But….Look at that loop….
- Flattened??

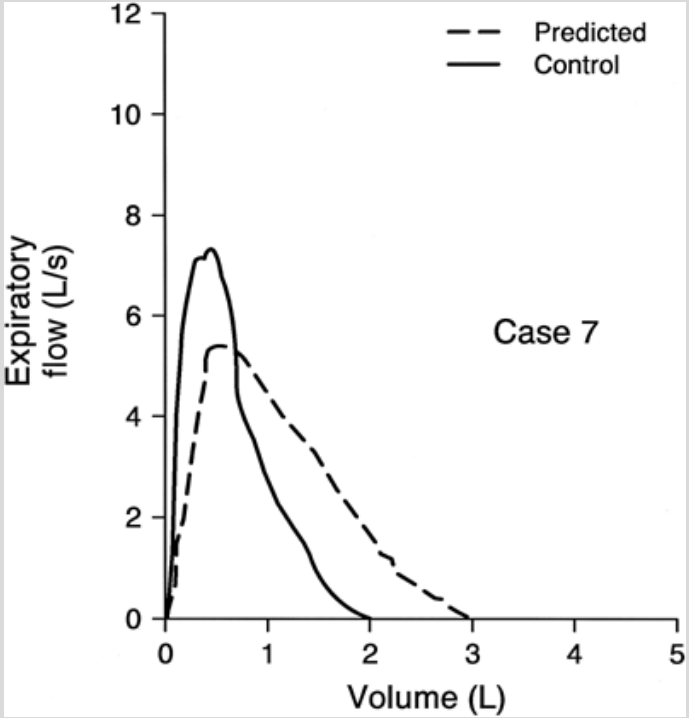
- **ANCA-associated Vasculitis**
- **Subglottic stenosis**

67 nonsmoker c occasional sob and wheezing; obese



- Nrml FEV1 ratio
- FEV1 --- 72%
- Nrml TLC

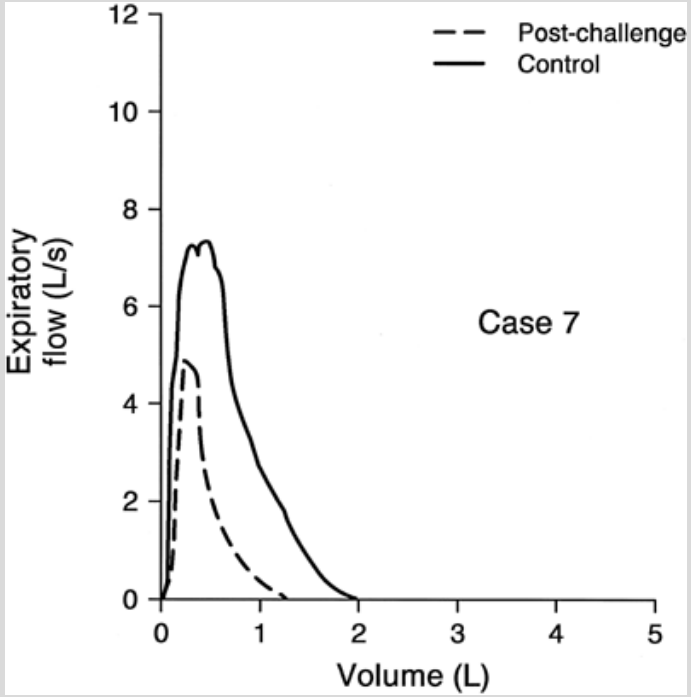
67 nonsmoker c occasional sob and wheezing; obese



- Nrml FEV1 ratio
- FEV1 72%
- Nrml TLC
- Looks restricted…
- Mild scoop..
- But wheezing, sob, so…..

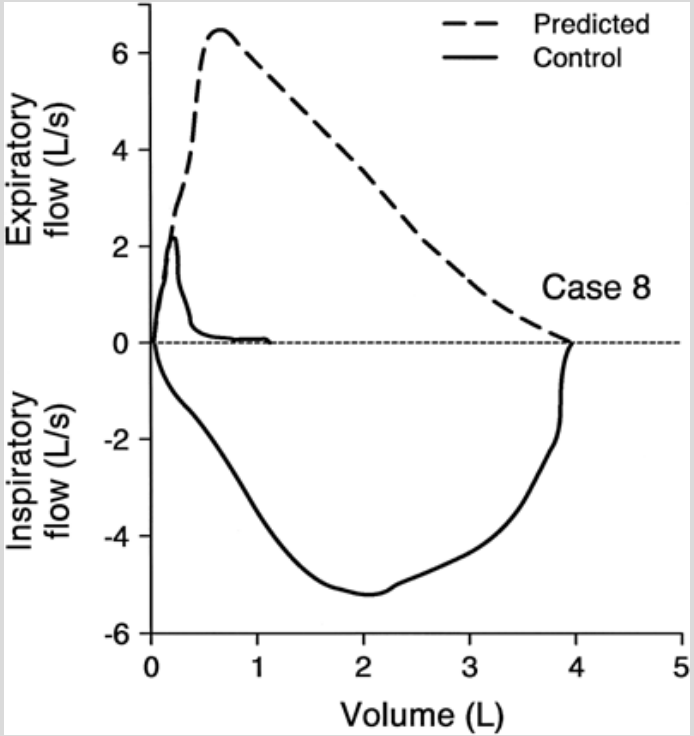
Methacholine Challenge!!!

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- Post-challenge 41% decrease in FEV1
- Treated c inhaled bronchodilator

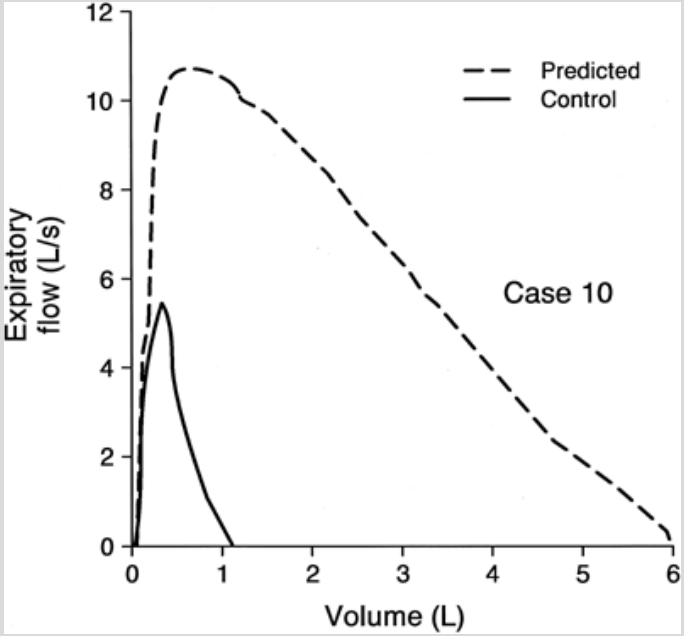
43 F 16 p-y smoker recent wheezing, CP, SOB



- Decreased FEV1 ratio
- FEV1 17%

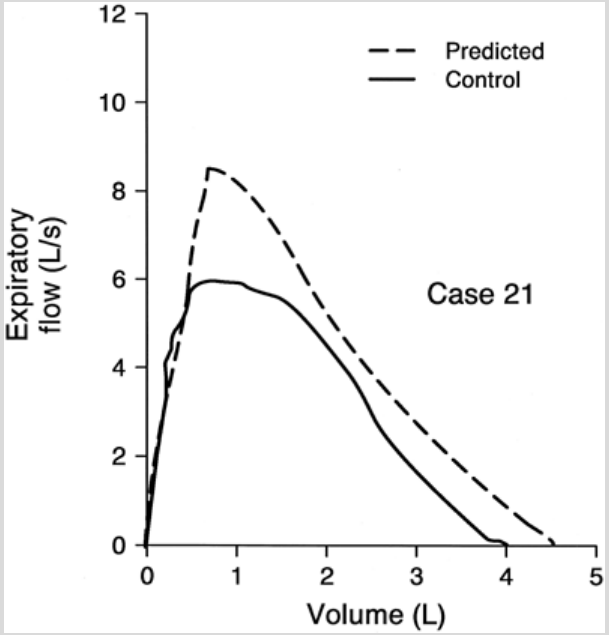
- Look at the loops….

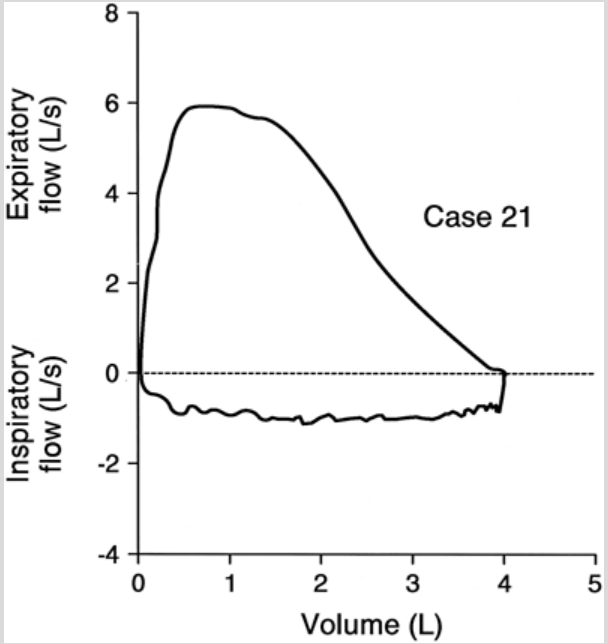
60 M SOB c exertion non-smoker



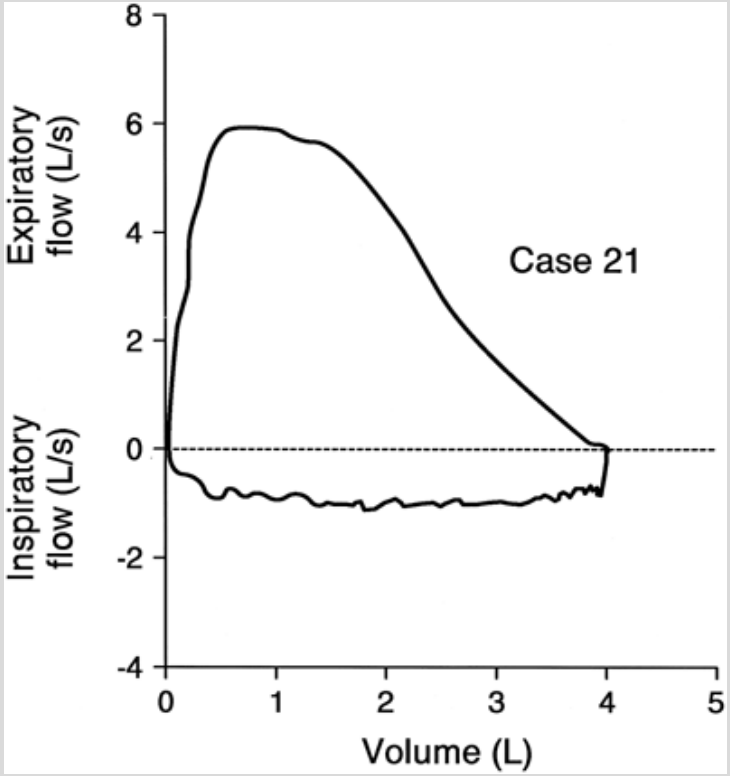
- FEV1 ratio Nrml
- TLC -- 28%
- FEV1 -- 21%
- FVC – 19%
- **Severe restrictive lung disease**

50 M – DOE nonsmoker
h/o bulbar poliomyelitis

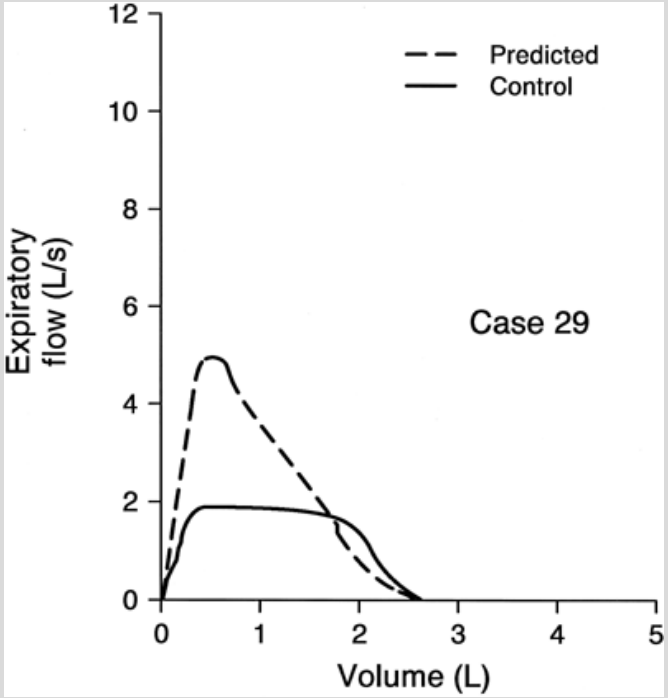




R Vocal cord paralysis
L partial vocal cord paralysis



64 F nonsmoker gradual onset DOE over 5 years, noisy breathing



- Where is the inspiratory loop???

Subglottic stenosis (and post-repair)

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